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Finding PRRSV in sow herds: Family oral fluids vs. serum samples from due-to-wean pigs

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ABSTRACT

The aim of this study was to compare the detection of porcine reproductive and respiratory syndrome virus (PRRSV) in due-to-wean litters in commercial swine breeding herds using family oral fluids (FOF) vs. individual piglet serum samples. FOF and piglet serum samples were collected in 199 due-to-wean litters on six farms containing 2177 piglets. All samples were individually tested for PRRSV RNA by RT-rtPCR. A litter was considered PRRSV-positive when PRRSV RNA was detected in ≥ 1 piglet serum sample or the FOF sample. Mixed effect logistic regression with farm as a random effect was used 1) to evaluate the probability of obtaining a PRRSV RNA positive FOF as a function of the proportion of viremic piglets in a litter and 2) the effect of litter size and parity on the probability that a litter would test PRRSV RNA positive in FOF. A Bayesian prevalence estimation under misclassification (BayesPEM) analysis was used to calculate the PRRSV prevalence and 95 % credible interval given the condition that all samples (FOF and serum) tested negative. In total, 34 of 199 litters (17.1 %) contained ≥ 1 viremic piglet(s), and 28 of 199 litters (14.1 %) were FOF positive. When all piglet serum samples within a litter tested negative, 1 of 165 FOF (0.6 %) tested PRRSV RNA positive. The probability of a PCR-positive FOF sample from litters with 10 %, 20 %, 30 %, 40 %, and 50 % within-litter PRRSV prevalence was 3.5 %, 35.1 %, 88.8 %, 99.2 %, and >99.9 %, respectively. The odds of a PCR-positive FOF in a first parity litter were 3.36 times (95 % CI: 2.10–5.38) that of a parity ≥ 2 litter. The odds of a positive FOF result in a litter with ≤ 11 piglets were 9.90 times (95 % CI: 4.62–21.22) that of a litter with > 11 piglets. FOF was shown to be an efficacious sample type for PRRSV detection in farrowing rooms. A risk-based approach for litter selection combined with FOF collection can be used to improve on-farm PRRSV detection with a limited sample size, compared to sampling multiple individual pigs. Finally, the BayesPEM analysis showed that PRRSV may still be present in breeding herds when all samples (serum and FOF) test PRRSV RNA negative, i.e., negative surveillance results should be interpreted with caution.

1. Introduction

In PRRSV elimination programs based on mass exposure of the sow herd to live wild-type virus or modified-live virus vaccine, the prevalence of infectious breeding females declines over time as animals clear the infection (Torremorell et al., 2002, 2003; Corzo et al., 2010). Elimination is achieved when all animals are free of PRRSV and anti-PRRSV antibodies.

It has been proposed by the American Association of Swine Veterinarians in 2011 that breeding herd PRRSV classification should be based on clinical data coupled with PRRSV RT-rtPCR testing of serum samples from due-to-wean piglets. Thus, a PRRSV breeding herd was classified as "positive stable" after four consecutive negative tests on due-to-wean piglet sera (n = 30) collected every 30 days (or more frequently) over a 90-day period. A minimum of 30 serum samples was required at each sampling in order to provide 95 % confidence to detect a prevalence ≥ 10 % (Holtkamp et al., 2011). A shortcoming of this approach is that PRRSV can persist in breeding herds at a prevalence of < 10 %. A longitudinal study of 56 herds detected a virus that matched the open reading frame (ORF)-5 sequence of the "original" herd virus in 5 herds (8.9 %) after failing to detect PRRSV in serum samples from pre-weaning piglets over 90 days (Linhares et al., 2014). In other studies, 3 of 825 sows (0.4 %) were PRRSV PCR positive on serum 15 months after the initiation of a PRRSV test-and-removal elimination project (Dee et al.,

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2000b), and PRRSV was detected by virus isolation and immunohistochemistry in 1 of 60 sows (1.7 %) two years after the initial PRRSV outbreak and 6-months after herd closure (Bierk et al., 2001). Altogether, these studies demonstrate that PRRSV can sustain low-prevalence infection in breeding herds undergoing virus elimination and illustrate the need for improved surveillance methods.

The collection of oral fluids for PRRSV diagnostics and surveillance was described by Prickett et al. in 2008 and subsequently researched extensively in growing/finishing pigs and adult animals (Prickett et al., 2008a, b; Kittawornrat et al., 2010, 2012; Ramirez et al., 2012; Kittawornrat et al., 2013; Olsen et al., 2013; Decorte et al., 2014; Kuiek et al., 2015; Biernacka et al., 2016; De Regge and Cay, 2016; Rotolo et al., 2017). With a variable degree of success rate across farms, oral fluids can be collected from due-to-wean litters. In contrast, a consistently high success rate has been reported for the collection of family oral fluids (FOF) from suckling litters before weaning (Almeida et al., 2020). The aim of this study was to compare the ability to detect PRRSV-infected litters using individual piglet serum samples vs. FOF in commercial swine breeding herds in the United States.

2. Material and methods

2.1. Study design

This study was designed to compare the probability of detecting PRRSV RNA in FOF versus individual due-to-wean piglet serum samples. FOF were collected from 199 litters containing 2177 piglets in 11 rooms on six farms (Farm A – 59 samples in three rooms; Farm B – 34 samples in one room; Farm C – 23 samples in three rooms; Farm D – 25 samples in one room; Farm E – 13 samples in one room; Farm F – 45 samples in one room). For all farms, farrowing rooms with the litters closest to weaning were selected for sampling. Individual serum samples were collected from piglets in litters where FOF was collected. This study was approved by the Iowa State University Office for Responsible Research under protocol #4-16-8240-S.

2.2. Sample collection

The study was conducted in conveniently selected farms located in Iowa and Illinois (Table 1). The criteria to enroll farms was allowing access to investigators for sample collection in due-to-wean litters, and breeding herds classified as PRRSV-positive unstable, i.e., recent diagnostic evidence of viral circulation in suckling pigs (Holtkamp et al., 2011). All samples were collected between May 2017 and September 2019. FOF samples were collected by hanging a length of 5/8-inch cotton rope (Skydog Rigging Equipment, Lake in the Hills, IL, USA) in a location that allowed access to both the sow and her piglets. Ropes were placed between 6:00 am and 7:00 am and fastened securely to farrowing crate bars using plastic ties. To encourage piglets to chew on the rope, it was placed so that the three unraveled strands hung approximately 2.54 cm (one inch) from the floor. Ropes were left in place for at least 30 min, after which oral fluids were harvested by placing the wet end in a plastic bag and then pulling the rope from the bag through clenched fingers. The fluid that pooled in the bag was then poured into a 50-mL conical centrifuge tube (Thermo Fisher Scientific, Waltham, MA, USA) for transport and storage. Clean disposable gloves and plastic bags were used for each FOF collection to avoid cross contamination between litters. Blood samples were collected from all piglets in litters from which FOF samples were collected using a single-use sterile BD Vacutainer™ SST™ Venous Blood Collection Tube (Thermo Fisher Scientific) via jugular venipuncture.

2.3. Sample handling

Immediately following collection, samples were labeled with key information: farm identification, collection date, and farrowing crate information (position within a farrowing room). Collected samples were maintained at 4° to 8 °C and transported to the diagnostic laboratory for processing within 24 h. In the laboratory, whole blood was centrifuged for 10 min at 1600 x *g* and then serum samples were submitted for PRRSV RNA testing.

2.4. Diagnostic testing

Serum and FOF samples were tested by PRRSV real-time reverse-transcriptase polymerase chain reaction (RT-rtPCR) at the Iowa State Veterinary Diagnostic Laboratory, Ames, IA using routine procedures. In brief, nucleic acids were extracted using a KingFisher® Flex automated magnetic particle processor system (Thermo Fisher Scientific) and a commercial extraction kit MagMAX™ Pathogen RNA/DNA Kit (Thermo Fisher Scientific) according to the manufacture's specifications. Samples were extracted using the high-volume chemistry. The high-volume lysis for PRRSV extraction contained: 100 μL of sample, 120 μL of lysis/binding solution concentrate, 120 μL of 100 % isopropanol, 2 μL of carrier RNA (1 μg/μL), and 2 μL of Xeno™ RNA (10,000 copies/μL) (Thermo Fisher Scientific). The high-volume wash steps used 300 μL of wash buffer 1 and 450 μL of wash buffer 2. Thereafter, the PRRSV RT-rtPCR was performed using the 10X PRRSV Primer Probe Mix V2 from the VetMAX™ PRRSV NA and EU kit (Thermo Fisher Scientific). The assay was modified from the original kit to use TaqMan® Fast Virus 1-Step Master Mix (4X) (Thermo Fisher Scientific) with the addition of Amplitaq 360 DNA Polymerase (Thermo Fisher Scientific). Each

Table 1
Relevant characteristics of sow herds participating in the study and number of individual piglet serum samples collected in each farm and results from PRRSV RT-rtPCR testing.

Farm	Sows (n)	Farrowing	PRRSV unstable	Most recent PRRSV outbreak	Immunologic intervention	Sampled litters (n)	Sampled piglets (n)		
							Negative	Positive	Total
A	1,100	Continuous	Yes	12 months	PRRSV killed vaccine[1]	59	430	134	564
B	6,500	Continuous	Yes	6 months	PRRS LVI[2]	34	372	0	372
C	350	2-week batch farrow	Yes	8 months	PRRS MLV[4]	23	243	12	255
D	3,000	Continuous	Yes	18 months	PRRSV MLV[3] quarterly in reproductive females	25	302	0	302
E	1,200	2-week batch farrow	Yes	9 months	Sows: LVI[2] Gilts: PRRSV killed vaccine[1] + LVI[2]	13	97	38	135
F	3,600	Continuous	Yes	6 months	LVI[2]	45	549	0	681
Total						199	1,993	184	2177

[1] MJPRRS Vaccine, Phibro Animal Health Corporation (Teaneck, NJ, USA).
[2] LVI - live virus inoculation.
[3] Fostera PRRS, Zoetis Animal Health (Parsippany-Troy Hills, NJ, USA).
[4] Ingelvac PRRS® MLV (St. Joseph, MO, USA).

reaction consisted of 6.5 μL of TaqMan® Fast Virus 1-Step Master Mix (4X), 0.8 μL Amplitaq 360 DNA Polymerase (5U/μL), 2.7 μL of nuclease-free water, 2.0 μL of the 10X PRRSV Primer Probe Mix V2, and 8.0 μL of nucleic acid template. Each plate included a positive extraction control, positive amplification control, negative extraction control, and negative amplification control. The assay was conducted on the ABI-7500 Fast system (Thermo Fisher Scientific), using the 7500 Fast System SDS Software (Version 1.4.0.27). The ABI-7500 was set to run in fast mode, with cycling conditions: 5 min at 50 °C; 20 s at 95 °C; 40 cycles of 95 °C for 3 s and 60 °C for 30 s.

2.5. Statistical analysis

Descriptive statistics (including distributions of frequency counts of sample sizes and test results, agreement between FOF and serum samples) were calculated using Microsoft Excel for Mac version 13.41 (2020). Statistical modeling analyses were conducted using R software version 3.5.2. (R Core Team, 2018). For all analyses, a litter was considered PRRSV-positive when PRRSV RNA was detected by RT-rtPCR in ≥ 1 piglet serum sample from the litter or the FOF sample from the litter. Mixed effect logistic regression with farm as a random effect was used to evaluate the probability of obtaining a PRRSV RNA positive FOF as a function of the proportion of viremic piglets in a litter. To compare detection by FOF vs. randomly selected piglet serum samples, the power of detection (POD) with n serum samples was calculated as $POD = 1 - (1 - \pi)^n$, where π is the within-litter disease prevalence determined by positive piglet serum samples. Mixed effect logistic regression was also used to evaluate the effect of litter size and parity on the probability that a litter would test PRRSV positive in FOF. In these analyses, parity was considered binary (1 = first parity, 0 = not first parity) and litter size was treated as a categorical variable, i.e., litters with ≤ 11 piglets were defined as small and litters > 11 piglets defined as large. The selection of this cutoff is data-drive. Under such a cutoff, the difference in the proportion of FOF positive litters between small and large sizes is maximum.

To estimate the prevalence in a piglet population in which all samples tested negative with either serum or FOF samples, a Bayesian prevalence estimation under misclassification method was used (Flor et al., 2020). Assuming serum and FOF test diagnostic sensitivity and specificity as se and sp, respectively, disease prevalence as π_0, and litter size as k, it can be derived that the probability of a PCR-positive test result (p) may be calculated for serum and FOF samples as given in Eqs. 1 and 2, respectively.

$$p\ (\text{serum}) = se \times \pi_0 + (1 - sp) \times (1 - \pi_0) \quad (1)$$

$$p\ (\text{FOF}) = se \times (1 - (1 - \pi_0)^k) + (1 - sp) \times (1 - \pi_0)^k \quad (2)$$

Derivation of Eqs. 1 and 2 can be found in the Appendix A. For simple random sampling from the population, it was assumed that the number of positive serum (or FOF) samples followed a Binomial distribution: $x \sim \text{Binomial}(n, p)$, where n is the serum (or FOF) sample size, and p can be replaced with Eq. 1 (or 2). The Bayesian method provided a way for updating prior beliefs about the unknown parameters to posterior beliefs based on the data. In this Bayesian framework, we observed the sample size n and the number of samples tested positive x, and aimed to estimate the unknown prevalence parameter π_0. The test sensitivity and specificity were assigned with priors of Beta distributions based on data from this study. A non-informative prior distribution Beta (1, 1) was chosed for π_0 to reflect our *a priori* lack of knowledge about it. Markov Chain Monte Carlo (MCMC) was conducted with R JAGS to obtain point and interval estimations for the prevalence (Plummer, 2003). For the MCMC, three chains with 10,000 iterations and 50 % burn-in session were performed. Finally, prevalence estimates (95 % CI) were calculated over a range of serum and oral fluid sample sizes for the condition in which all samples tested PRRSV RNA negative.

3. Results

A total of 2177 individual piglet blood samples and matching family oral fluid samples were collected from 199 litters in six farms (Table 1). Family oral fluid sampling was unsuccessful in an additional 18 litters (92.7 % success rate). Based on PRRSV RNA testing, 34 of 199 litters (17.1 %) contained ≥ 1 viremic piglet(s) and 28 of 199 litters (14.1 %) were FOF-positive. Among the 165 litters in which all piglet serum samples tested negative, one FOF (0.6 %) tested PRRSV RNA positive. Thus, 164 litters (82.4 %) were PRRSV-negative by both FOF and individual piglet serum samples (Table 2). The agreement level between the FOF and serum samples is fairly strong with the Cohen's kappa coefficients $\kappa = 0.868$. As shown in Table 3, 23 of 24 litters (95.8 %) with ≥ 3 viremic piglets were PRRSV RNA positive in FOF; in 10 litters with ≤ 2 viremic piglets, 5 (50.0 %) were FOF-positive. A mixed effects logistic regression analysis based on the data presented in Table 4 estimated the probability of a positive FOF sample from litters with 10 %, 20 %, 30 %, 40 %, and 50 % within litter PRRSV prevalence as 3.5 %, 35.1 %, 88.8 %, 99.2 %, and >99.9 %, respectively (Fig. 1). The probability of detecting PRRSV RNA in family oral fluids was ≥ 95 % when the within-litter prevalence was ≥ 33.3.%. Detection of positive litters using piglet serum samples was highly dependent on within-litter prevalence. That is, achieving ≥ 95 % probability of detection with 1, 2, 3, 4, or 5 piglet serum samples would require within-litter prevalence of 95.0 %, 77.6 %, 63.1 %, 52.6 %, and 45.0 %, respectively (Fig. 1).

Holding litter size constant, the odds of obtaining a PCR-positive FOF of a first parity litter was 3.36 times (95 % CI: 2.10–5.38, p-value < 0.001) that of a parity ≥ 2 litter. Holding parity constant, the odds of a positive FOF result for litters with ≤ 11 piglets was 9.91 times (95 % CI: 4.62–21.22, p-value < 0.001) that of a litter with > 11 piglets. The results of the Bayesian prevalence estimation under misclassification analysis for the condition in which all samples (serum and FOF) test PRRSV RNA negative are presented in Table 4.

4. Discussion

PRRSV has caused significant economic and welfare losses in the global swine industry for over 30 years. During that time, various strategies to control and eliminate the virus from sow herds have been proposed and implemented; none of which is perfect. Regardless of the elimination strategy employed, it is critical is to accurately determine the true status of virus circulation in the herd. In 2011, the American Association of Swine Veterinarians (AASV) PRRS Task Force described standardized nomenclature and sampling/testing procedures for establishing sow herd PRRSV status (Holtkamp et al., 2011). Widely adopted by producers and veterinarians, the availability of standardized procedures and terminology have facilitated communication concerning PRRSV control and elimination.

The 2011 AASV guidelines were based on the assumption that the PRRSV status of piglets at weaning is a proxy for the PRRSV status of the sow herd. Using a classical probability-based sampling approach (Cannon and Roe, 1982), the protocol recommended monthly testing ≥ 30 piglet sera. Today, farms attempting to produce PRRSV-negative piglets after an outbreak continue to use this testing strategy to determine when stability is achieved, as defined by negative PRRSV RT-rtPCR testing results over 4 consecutive samplings in a 90-day period (Holtkamp et al.,

Table 2
Litter level PRRSV status agreement between family oral fluid (FOF) samples and serum samples.

		Serum		Total (%)
		Negative (%)	Positive (%)	
FOF	Negative (%)	164 (82.4)	6 (3.0)	170 (85.4)
	Positive (%)	1 (0.5)	28 (14.1)	29 (14.6)
	Total (%)	165 (82.9)	34 (17.1)	199 (100.0)

Table 3
Family oral fluid (FOF) PRRSV PCR results according to the number of viremic piglets within a litter as determined by PRRSV PCR on individual serum samples.

FOF PRRSV PCR results	Count of litters by number of PRRSV viremic piglets within a litter determined by individual serum PCR												Total
	0	1	2	3	4	5	6	7	8	9	10	11	
Negative	164	4	1	0	1	0	0	0	0	0	0	0	170
Positive	1	4	1	1	4	1	2	3	5	1	4	2	29
TOTAL	165	8	2	1	5	1	2	3	5	1	4	2	199

Table 4
PRRSV prevalence 95 % credible interval given that all samples (serum and family oral fluids (FOF)) test PRRSV RNA negative as calculated from a Bayesian prevalence estimation under misclassification (BayesPEM) analysis.

Serum samples (n)	FOF samples (n)	95 % credible interval - lower and upper boundaries
12	4	0 to ~20.0
30	5	0 to ~9.0
60	8	0 to ~5.0
90	11	0 to ~3.0
120	15	0 to ~2.0
240	30	0 to ~1.0
400	40	0 to ~0.5

2011). Under the assumption of hypergeometric distribution, this protocol provided 95 % confidence of finding ≥ 1 positive animal(s) in a large (>1000 pigs) population with a prevalence of ≥ 10 %.

Producers often assume that achieving PRRSV stability implies that the virus is no longer being shed and transmitted within the sow herd. Not infrequently, producers then relax biosecurity protocols, begin the reintroduction of PRRSV-naïve gilts, and return to cross-fostering within and between farrowing rooms. The introduction of PRRSV-naïve animals provides the opportunity for PRRSV to re-establish itself in the herd when prevalence is non-zero but undetected. Evidence of this phenomenon is the detection of viruses consistent with historic herd isolates, as determined by ORF5 sequencing, in PRRSV stable herds (Linhares et al., 2014). On the whole, this implies that PRRSV can sustain low prevalence in sow herds over time and suggests that currently accepted protocols are not adequate for detecting the virus under low prevalence conditions (Dee et al., 2000a; Bierk et al., 2001; Linhares et al., 2014).

An approach to PRRSV detection at prevalence < 10 % is needed to avoid misclassifying breeding herd PRRSV status. Increasing individual piglet serum sample size is one option, but the number of piglets that would need to be bled for detection at low prevalence is prohibitively large and a challenge for producer's fiscal and labor resources. As demonstrated by Salman (2003), as prevalence approaches zero, sample sizes required for detection quickly increase toward the total population size. Alternatively, oral fluids are an aggregate sample that, at a group/herd level, provides a higher likelihood of pathogen detection with less producer effort and at a lower cost than serum samples (Henao-Diaz et al., 2020). Oral fluid collection, done by hanging a cotton rope in a farrowing crate that is only accessible to the piglets, has not been repeatable in due-to-wean piglet litters in commercial herds, and therefore, it has not been widely adopted by producers. Family oral fluid collection, in turn, is an adaptation that takes advantage of the piglet's natural tendency to mimic their mother's behavior, i.e., as the dam interacts with the rope, her piglets do, as well (Yeske-Livermore et al., 2014; Almeida et al., 2020). Almeida et al. (2020) reported a success rate of ≥ 91.5 % (183 of 200 litters attempted) using this approach.

Although family oral fluid samples could provide the means to increase the number of piglets represented in the sample, there is limited data on the detection of PRRSV in this matrix. Moreover, there is even less data on the herd sensitivity of FOF to detect PRRSV RNA in the suckling pig population. The results of this study demonstrated that the litter level agreement between serum (199 positive litters) and family oral fluid (193 positive litters) samples was high (96.5 %). The probability of PRRSV detection in family oral fluids was dependent on the proportion of viremic piglets in the litter (logistic regression), but detection probabilities for family oral fluids were generally higher than

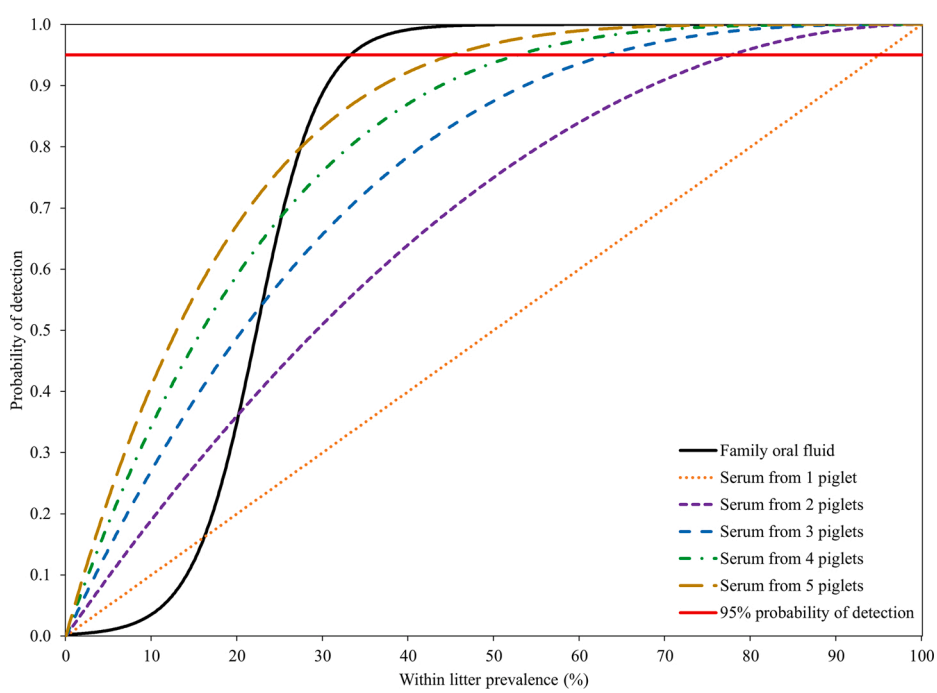

Fig. 1. Probability of PRRSV RNA detection using family oral fluids (FOF) according to number of viremic piglets within a litter.

detection based on piglet serum. Detection based on sampling individual piglets is particularly problematic when, as in the present study, more than 50 % of positive litters had ≤ 5 viremic piglets. In the field, one piglet serum sample is usually collected from litters, an approach that will decrease the probability of PRRSV detection at the litter level. The detection of one FOF sample as PRRSV RNA positive in the absence of viremic piglets was not further investigated. PRRSV shedding by the sow or environmental contamination are possible explanations for this finding.

Overall, the results of this study are consistent with previous reports of population-based monitoring protocols. Lopez et al. (2020) found similar results when evaluating the probability of PRRSV detection in litters by processing fluids compared to individual piglet serum. Olsen et al. (2013) reported a similar trend between in-pen PRRSV prevalence and the probability of PRRSV RNA or antibody detection in pen-based oral fluids from growing pigs. Lebret et al. (2019) compared PRRSV RNA detection based on one piglet serum sample vs. oral fluid from the same litter and reported results of 62 % vs. 77 % for serum and oral fluids, respectively.

An issue separate from, but related to, declining prevalence is the clustering (non-random distribution) of positive litters within farrowing rooms. In the presence of clustering, random sampling is inefficient and requires a large sample size to achieve detection (Almeida, 2020). One way to approach this issue is to focus on sampling high-risk litters. In this study, high-risk litters were identified as first parity sows and sows of any parity with small litters (≤ 11 piglets). Specifically, the odds of a positive family oral fluid in first parity sow litters were 3.4 times (95 % CI: 2.1–5.4) that of parity ≥ 2 sows, and the odds small litters were 9.9 times (95 % CI: 4.6–21.2) that of litters with > 11. Relatively small litter size and first parity sows have previously been reported to be associated with a higher risk of PRRSV and other viral infections (Dee and Joo, 1994; Calsamiglia and Pijoan, 2000; Sasseville et al., 2001; Alsop, 2006; Quiroga et al., 2008; Truyen and Streck, 2019).

Sample size is invariably a point of dynamic tension because of the conflicting goals of acquiring the information and avoiding unnecessary costs. In the best cases, a surveillance sample size is selected to detect the virus at or above a given prevalence (Cannon and Roe, 1982; Holtkamp et al., 2011). Regardless, it is not uncommon for veterinarians and producers to conclude that the target pathogen is no longer present in the population when receiving negative test results (Salman, 2003). Of course, negative testing results for a given sample size actually mean that the target pathogen was not detected in the population at the target prevalence with a designated level of confidence. The Bayesian prevalence estimation under misclassification analysis presented in this study expanded upon this concept and provided estimates of the range of PRRSV prevalence possible given negative test results. Ultimately, it is essential to keep in mind the limitations of the sample numbers and the higher confidence that comes with increasing the sample size and using aggregate specimens.

5. Conclusions and implications

Family oral fluids were shown to be an efficacious sample type for PRRSV detection in farrowing rooms. A risk-based approach for litter selection combined with family oral fluid collection can improve on-farm PRRSV detection with fewer samples than required for piglet serum-based surveillance. In this study, parity and litter size were both identified as significant risk indicators for PRRSV RNA detection in family oral fluids and may be used to guide sample selection. Finally, data presented herein suggests that producers and veterinarians should not over-interpret negative results. In other words, even though negative results are encouraging for herds attempting to clear virus circulation, negative results should be interpreted in the context of the power of detection provided by the sample size used.

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Declaration of Competing Interest

The authors report no conflict of interest.

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Appendix A

Below is the derivation of Eqs. 1 and 2, the probability of a PCR-positive test result (p) for serum and FOF samples.

Denote $P(\cdot)$ as the generic probability of an event. Let u and v be the test result and the real disease status, respectively. Also, let "+" and "−" in the superscript represent being positive and negative, respectively. Therefore, the probability of a PCR−positve test result for serum or FOF samples can be written as

$$p = P(u^+v^+)\cdot P(v^+) + P(u^+v^-)\cdot P(v^-) = P(u^+v^+)\cdot P(v^+) + [1 - P(u^-v^-)]\cdot[1 - P(v^+)]$$

According to the definition of sensitivity and specificity, $se = P(u^+v^+)$, and $sp = P(u^-v^-)$. $P(v^+)$ is different for the two sample types. For serum samples, it is simply π_0; for FOF samples, it is $1 - (1 - \pi_0)^k$. Hence, the result follows.

$$p\ (\text{serum}) = se \times \pi_0 + (1 - sp) \times (1 - \pi_0) \tag{1}$$

$$p\ (\text{FOF}) = se \times (1 - (1 - \pi_0)^k) + (1 - sp) \times (1 - \pi_0)^k \tag{2}$$

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